

DEPREDICT

Polymarket Backtest Report

Can AI Multi-Expert Deliberation Generate Alpha
in Prediction Markets?

48 Resolved Predictions · 38 Non-Crypto · 10 Crypto

Period: 2026-03-19 → 2026-03-23

Strategy: Kelly Criterion (Half-Kelly) · Initial Bankroll: \$1,000

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github.com/Andrewyzzz/Depredict

Executive Summary

Overall ROI	Non-Crypto ROI	Crypto ROI	Win Rate (All)	Brier Edge	Brier Wins
+131.2%	+245.5%	-33.1%	54.2%	+0.0037	26/48

Category	Trades	Win Rate	ROI	P&L	Sharpe	Max DD	Final
All	48	54.2%	+131.2%	\$+1,312	+0.19	71.1%	\$2,312
Non-Crypto	38	60.5%	+245.5%	\$+2,455	+0.27	61.0%	\$3,455
Crypto	10	30.0%	-33.1%	\$-331	-0.08	52.3%	\$669

KEY FINDING: DePredict generates significant alpha on non-crypto markets (+245% ROI, 61% win rate) but consistently loses on crypto predictions (-33% ROI). Optimal strategy: deploy capital exclusively on non-crypto markets with half-Kelly sizing.

Methodology — System & Signal

1. DePredict Multi-Expert Deliberation System

DePredict assembles 10 AI expert agents, each with a distinct analytical stance (bull, bear, neutral) and domain-specific knowledge profile, to deliberate on binary prediction market questions. Rather than relying on a single model's output, the system creates 'structured cognitive conflict' — forcing experts to surface blind spots, challenge each other's reasoning, and converge toward calibrated forecasts.

The 10 agents span diverse analytical perspectives:

- Optimistic Analyst (Bull)
- Bayesian Analyst (Neutral)
- Data Statistician (Neutral)
- Contrarian Investor (Bear)
- Risk Assessor (Bear)
- Skeptical Analyst (Bear)
- Historical Analogy Analyst (Neutral)
- Sentiment Analyst (Bull)
- Fundamentals Analyst (Neutral)
- Synthesis Strategist (Neutral)

2. Three-Round Debate Pipeline

Round 1 — Independent Prediction: Each agent receives RAG-retrieved documents (YouTube transcripts + Tavily news) and produces an independent probability estimate (0–100%) with reasoning. Information is partitioned: 40% shared docs + 60% private docs per agent, creating genuine information asymmetry.

Round 2 — Cross-Rebuttal: Agents see all Round 1 predictions and reasoning, then write rebuttals challenging specific points. Agents may revise their probability in light of new arguments.

Round 3 — Final Prediction: After reading all rebuttals, each agent commits to a final probability, incorporating the strongest arguments from the debate.

3. Nine Aggregation Mechanisms

The 10 agents' final predictions are aggregated via 9 independent mechanisms:

- Simple Average — arithmetic mean of all agent probabilities
- Median — robust to outlier predictions
- Trimmed Mean — removes top/bottom 10% before averaging
- Logit Average — aggregation in log-odds space (handles extremes better)
- Extremized Average — pushes consensus away from 50% (Baron et al. 2014)
- Reputation-Weighted — weights by historical Brier Score performance
- LMSR Market — Hanson's Logarithmic Market Scoring Rule with Kelly sizing
- Peer Prediction (BTS) — Bayesian Truth Serum, scores via meta-predictions

Methodology — Trading & Position Sizing

4. Trading Signal Generation

Polymarket is a binary prediction market: YES shares trade at price $P_{\text{mkt}} \in (0,1)$, paying *1 if the event occurs and 0 otherwise*. The market price equals the implied probability. Our trading signal exploits discrepancies between DePredict's model probability P_{model} and the market price P_{mkt} .

Signal Logic:

If $P_{\text{model}} > P_{\text{mkt}} \rightarrow$ BUY YES (model thinks event is more likely than market)
If $P_{\text{model}} < P_{\text{mkt}} \rightarrow$ BUY NO (model thinks event is less likely than market)
Edge = $|P_{\text{model}} - P_{\text{mkt}}|$

P&L Calculation:

BUY YES at cost $P_{\text{mkt}} \rightarrow$ if YES resolves: profit = $\text{bet} \times (1 - P_{\text{mkt}}) / P_{\text{mkt}}$
 $\rightarrow$ if NO resolves: loss = $-\text{bet}$
BUY NO at cost $(1 - P_{\text{mkt}}) \rightarrow$ if NO resolves: profit = $\text{bet} \times P_{\text{mkt}} / (1 - P_{\text{mkt}})$
 $\rightarrow$ if YES resolves: loss = $-\text{bet}$

5. Kelly Criterion Position Sizing

The Kelly Criterion determines the optimal fraction of bankroll to wager, maximizing long-run geometric growth while avoiding ruin.

For a BUY YES trade (net odds $b = (1 - P_{\text{mkt}}) / P_{\text{mkt}}$):

$$f^* = \frac{P_{\text{model}} \times b - (1 - P_{\text{model}})}{b}$$

For a BUY NO trade (net odds $b = P_{\text{mkt}} / (1 - P_{\text{mkt}})$):

$$f^* = \frac{(1 - P_{\text{model}}) \times b - P_{\text{model}}}{b}$$

Where f^* is the fraction of bankroll to bet. If $f^* \leq 0$, skip the trade.

6. Half-Kelly & Risk Management

Full Kelly is theoretically optimal but assumes perfect probability estimates. In practice, model uncertainty means full Kelly over-bets, creating extreme volatility and drawdowns.

Methodology — Evaluation Framework

7. Profitability Metrics

- Total P&L — absolute dollar profit/loss from initial bankroll
- ROI (Return on Investment) — $\text{P\&L} / \text{initial_bankroll} \times 100\%$
- Win Rate — fraction of trades that result in positive P&L
- Sharpe Ratio — $\text{mean(per-trade return)} / \text{std(per-trade return)}$
Measures risk-adjusted returns; >0.5 is good, >1.0 is excellent
- Max Drawdown — largest peak-to-trough decline as % of peak bankroll
Measures worst-case capital erosion

8. Prediction Quality Metrics

- Brier Score — $(P_{\text{predicted}} - \text{outcome})^2$, where $\text{outcome} \in \{0, 1\}$
Ranges 0 (perfect) to 1 (worst); lower is better
- Model vs Market Brier — compares DePredict Brier to market-implied Brier
 $(\text{market_price} - \text{outcome})^2$; model wins if its Brier < market Brier
- Edge Distribution — histogram of $(P_{\text{model}} - P_{\text{mkt}})$ colored by W/L
Reveals whether large edges correlate with wins

9. Sensitivity Dimensions

- Kelly Fraction (0.1x → 1.0x) — find optimal risk level
- Edge Threshold (0% → 20%) — filter low-conviction signals
- Aggregation Mechanism (9 methods) — which aggregator is most profitable
- Category (Crypto vs Non-Crypto) — domain-specific performance

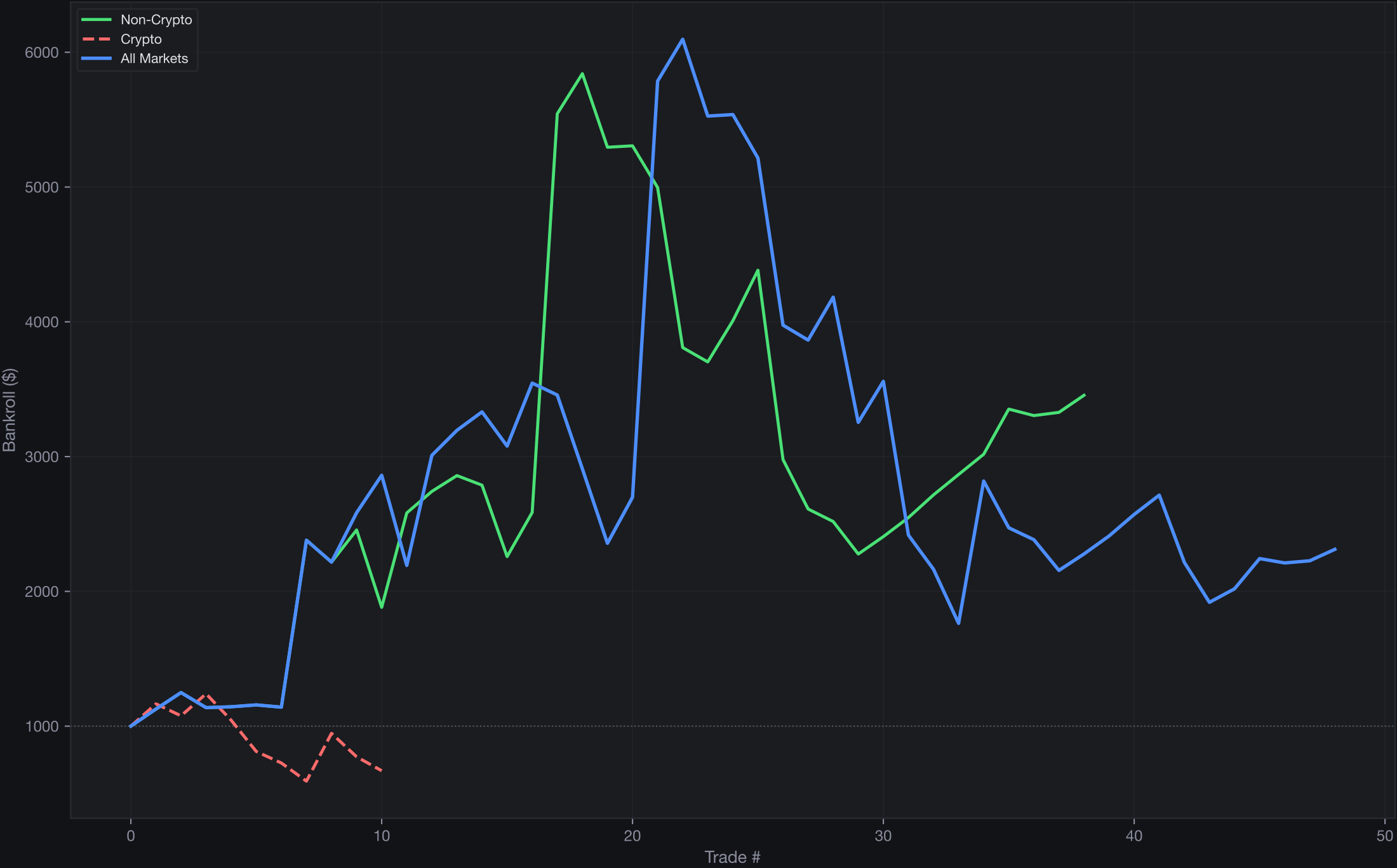
10. Backtest Assumptions & Constraints

- Execution at recorded market price (no slippage model)
- No transaction costs by default (Polymarket has no trading fees, but bid/ask spread exists — sensitivity can be run with --friction)
- Sequential trades, bankroll updated after each resolution
- No look-ahead bias: only `market_price_at_prediction` is used
- No position limits or liquidity constraints
- All predictions use the same Kelly multiplier (no adaptive sizing)

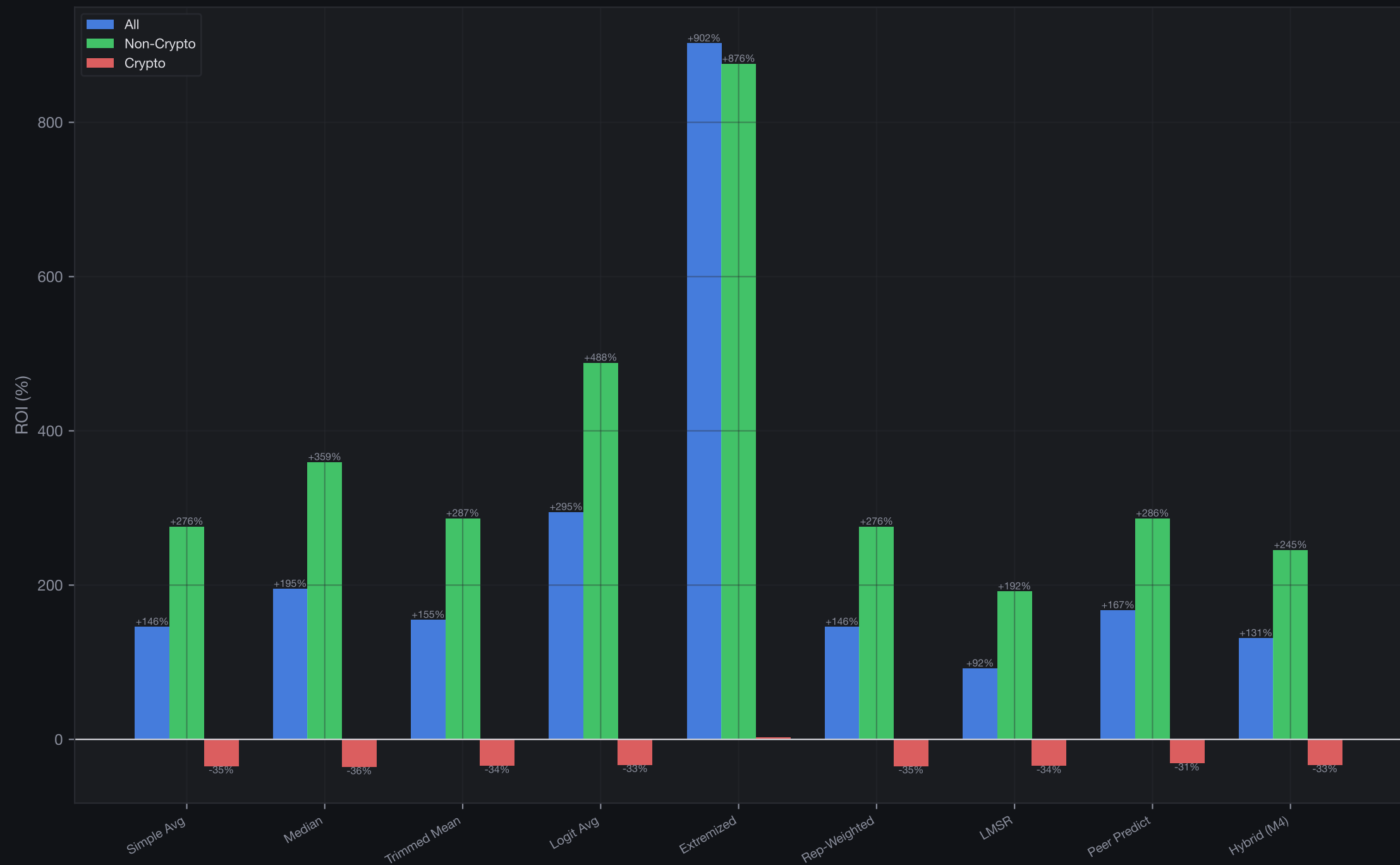
11. Data Collection

- Markets sourced via Polymarket API (`gamma-api.polymarket.com`)
- Pre-filtered: excluded extreme prices (<5% or >95%), derivative markets (props, spreads, O/U), geopolitical/violence topics
Resolution checked automatically via API polling

Equity Curve — Hybrid (Half-Kelly)

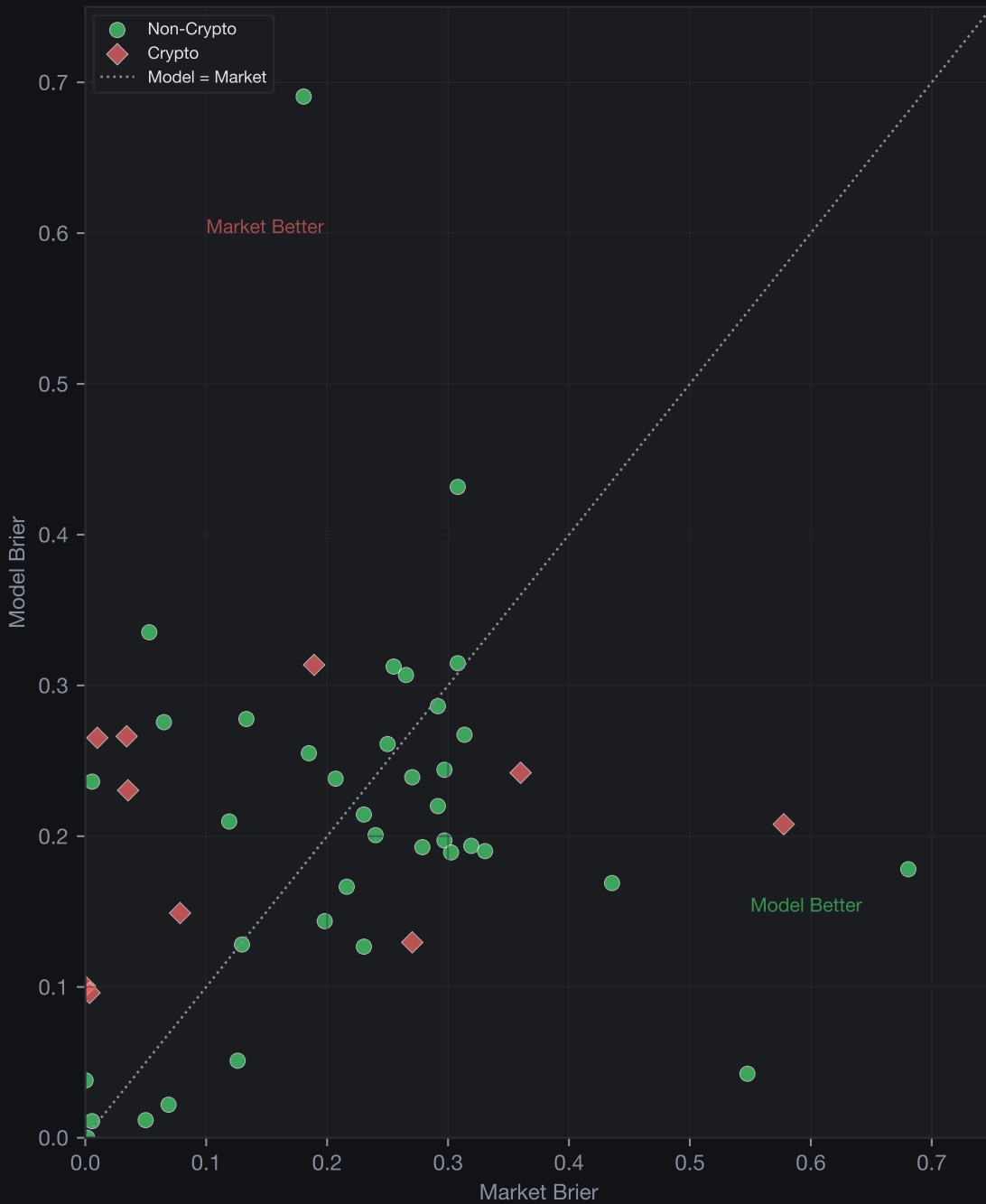


ROI by Aggregation Mechanism & Category

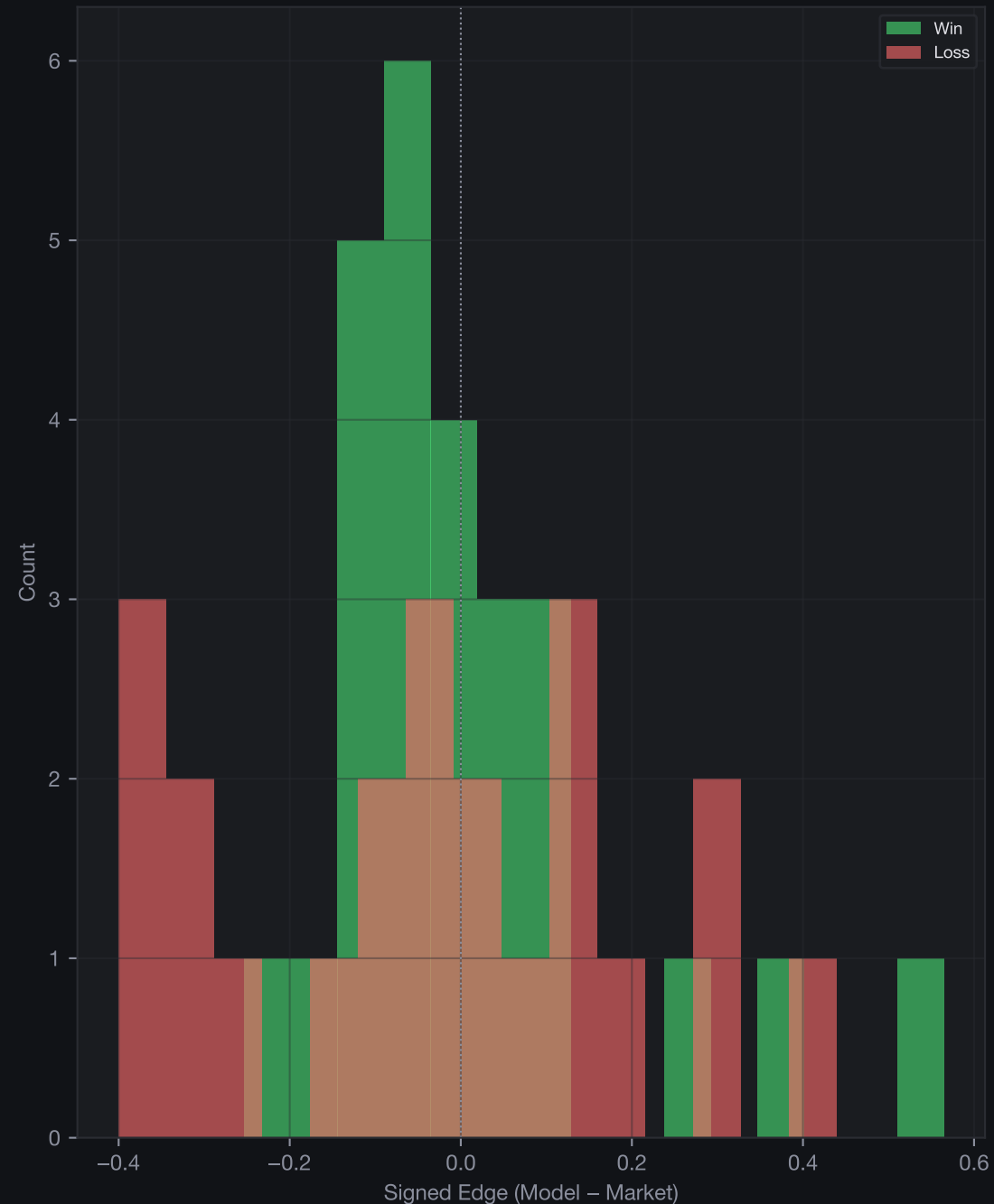


Prediction Quality — Brier Score & Edge Distribution

Model vs Market Brier Score



Edge Distribution by Outcome



Performance by Sub-Category

P&L by Sub-Category

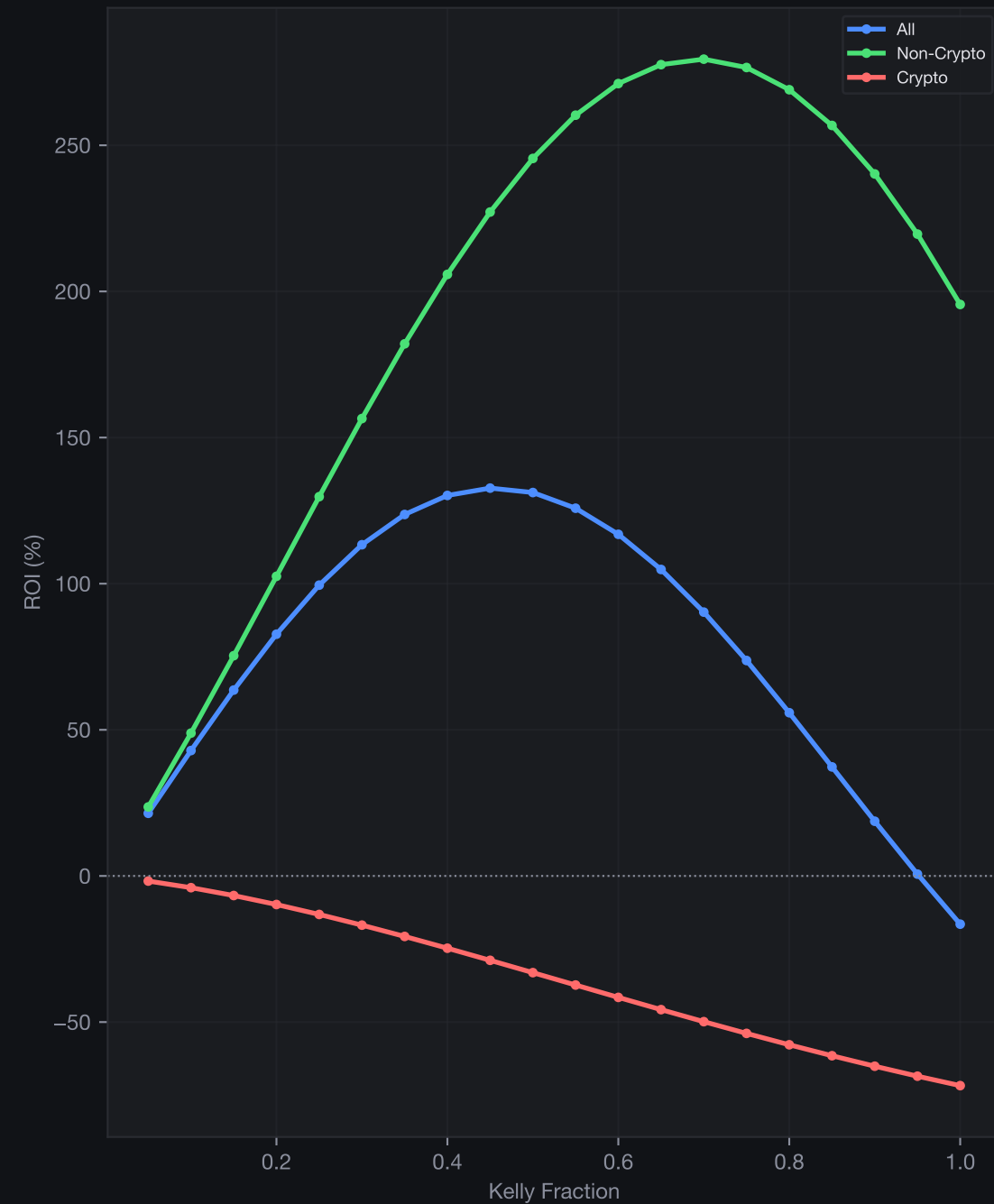


Win Rate by Sub-Category

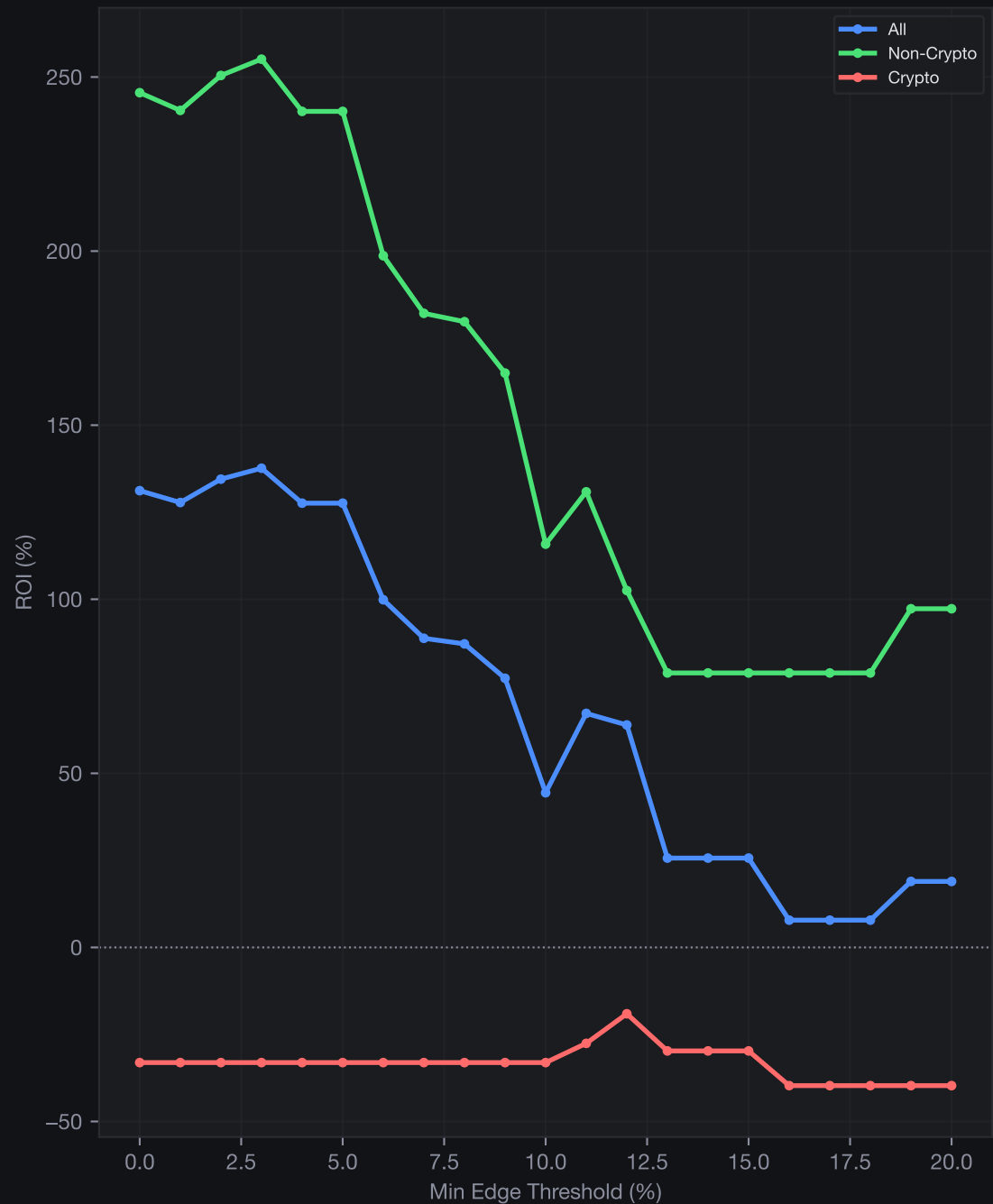


Sensitivity Analysis

Kelly Fraction vs ROI



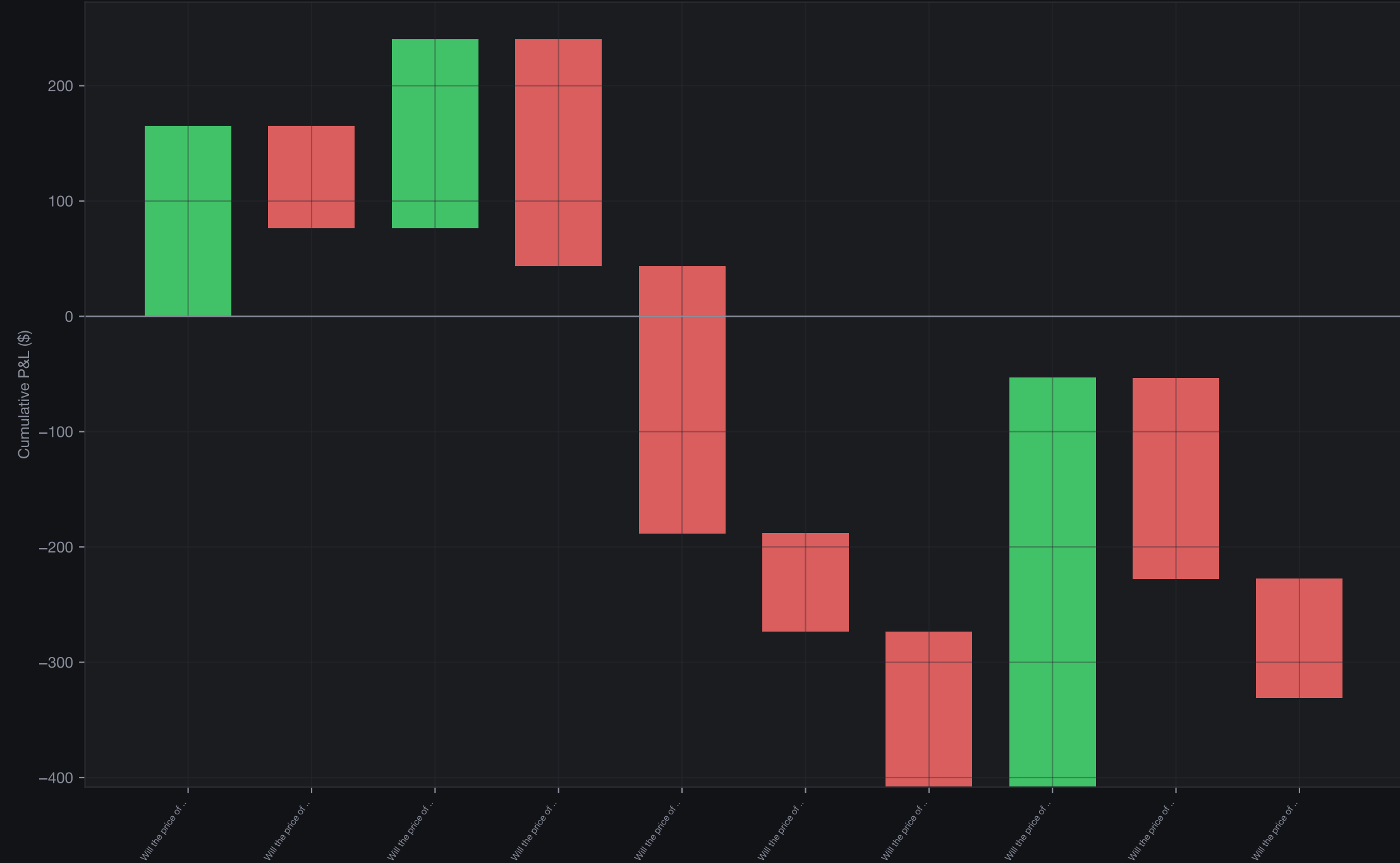
Edge Threshold vs ROI



Per-Trade P&L Waterfall — Non-Crypto



Per-Trade P&L Waterfall — Crypto



Trade Log — Non-Crypto (Page 1/2)

#	Question	Side	Model	Market	Edge	Kelly	Bet	P&L	Bank	W/L
1	Lakers vs. Rockets	YES	54.5%	43.5%	11.0%	9.8%	\$98	\$+127	\$1,127	W
2	Stars vs. Avalanche	YES	55.4%	45.5%	9.9%	9.1%	\$102	\$+122	\$1,249	W
3	South Florida Bulls vs. L...	YES	46.2%	34.5%	11.7%	8.9%	\$112	\$-112	\$1,137	L
4	Blues vs. Flames	NO	53.5%	54.0%	0.5%	0.5%	\$5	\$+6	\$1,144	W
5	Will Jesus Christ return ...	NO	1.5%	3.9%	2.3%	30.1%	\$345	\$+14	\$1,158	W
6	Counter-Strike: PARIVISIO...	NO	62.1%	64.0%	1.9%	1.4%	\$17	\$-17	\$1,141	L
7	Will AS Roma vs. Bologna ...	YES	82.5%	26.0%	56.5%	38.1%	\$435	\$+1,239	\$2,380	W
8	Counter-Strike: PARIVISIO...	NO	49.2%	57.0%	7.8%	6.8%	\$162	\$-162	\$2,217	L
9	Will AS Roma win on 2026-...	NO	45.3%	55.0%	9.7%	8.8%	\$195	\$+238	\$2,456	W
10	Game Handicap: GEN (-2.5)...	NO	41.0%	77.0%	36.0%	23.4%	\$574	\$-574	\$1,881	L
11	Bucks vs. Jazz	NO	40.7%	66.0%	25.3%	19.2%	\$361	\$+701	\$2,582	W
12	Utah vs. Golden Knights	YES	51.7%	46.0%	5.7%	5.3%	\$136	\$+159	\$2,742	W
13	Utah vs. Golden Knights: ...	NO	52.2%	56.0%	3.8%	3.4%	\$92	\$+117	\$2,859	W
14	Missouri Tigers vs. Miami...	YES	48.2%	45.5%	2.7%	2.5%	\$71	\$-71	\$2,788	L
15	Akron Zips vs. Texas Tech...	YES	53.8%	25.5%	28.3%	19.0%	\$530	\$-530	\$2,258	L
16	Lakers vs. Heat	YES	54.8%	42.5%	12.3%	10.7%	\$242	\$+328	\$2,585	W
17	LoL: G2 Esports vs BNK FE...	YES	57.6%	17.5%	40.1%	24.3%	\$628	\$+2,958	\$5,544	W
18	Will Manchester United FC...	NO	40.8%	46.5%	5.7%	6.2%	\$342	\$+297	\$5,841	W
19	Netanyahu out by March 31?	YES	20.0%	1.7%	18.4%	9.3%	\$546	\$-546	\$5,295	L
20	Vanderbilt Commodores vs...	NO	55.3%	55.5%	0.2%	0.2%	\$9	\$+11	\$5,306	W
21	Texas Tech Red Raiders vs...	YES	56.3%	50.5%	5.8%	5.8%	\$309	\$-309	\$4,997	L
22	Will "Project Hail Mary" ...	YES	51.5%	7.5%	44.0%	23.8%	\$1188	\$-1,188	\$3,809	L
23	Will "Project Hail Mary" ...	NO	87.3%	92.5%	5.2%	2.8%	\$106	\$-106	\$3,703	L
24	Will the highest temperat...	NO	45.0%	52.8%	7.8%	7.4%	\$273	\$+305	\$4,008	W
25	Will Elon Musk post 115-1...	NO	23.4%	35.5%	12.1%	17.0%	\$682	\$+375	\$4,383	W
26	Will Elon Musk post 90-11...	NO	20.6%	57.5%	36.9%	32.1%	\$1405	\$-1,405	\$2,978	L
27	Lakers vs. Pistons	YES	66.5%	55.5%	11.0%	12.3%	\$367	\$-367	\$2,611	L
28	Spread: Lakers (-1.5)	YES	54.9%	51.5%	3.4%	3.6%	\$93	\$-93	\$2,518	L

Trade Log — Non-Crypto (Page 2/2)

#	Question	Side	Model	Market	Edge	Kelly	Bet	P&L	Bank	W/L
29	Senators vs. Rangers	NO	51.3%	63.5%	12.2%	9.6%	\$241	\$-241	\$2,277	L
30	Senators vs. Rangers: 0/U..	NO	49.3%	54.5%	5.2%	4.7%	\$108	\$+129	\$2,406	W
31	Warriors vs. Mavericks	YES	62.0%	55.5%	6.5%	7.3%	\$176	\$+141	\$2,547	W
32	Will Elon Musk post 380-3..	NO	12.1%	22.4%	10.3%	23.0%	\$586	\$+169	\$2,716	W
33	Will Elon Musk post 400-4..	NO	18.0%	26.2%	8.2%	15.7%	\$425	\$+151	\$2,867	W
34	Spread: Lakers (-2.5)	NO	43.7%	49.0%	5.3%	5.4%	\$156	\$+150	\$3,017	W
35	Lakers vs. Pistons: 0/U 2..	NO	36.5%	48.0%	11.5%	12.0%	\$363	\$+335	\$3,353	W
36	Spread: Warriors (-2.5)	NO	48.6%	50.0%	1.4%	1.4%	\$48	\$-48	\$3,304	L
37	Warriors vs. Mavericks: 0..	YES	52.8%	52.0%	0.7%	0.8%	\$26	\$+24	\$3,328	W
38	Warriors vs. Mavericks: 0..	YES	51.7%	48.0%	3.7%	3.5%	\$117	\$+127	\$3,455	W

Trade Log — Crypto (Page 1/1)

#	Question	Side	Model	Market	Edge	Kelly	Bet	P&L	Bank	W/L
1	Will the price of Bitcoin..	NO	36.1%	52.0%	15.9%	15.3%	\$153	\$+165	\$1,165	W
2	Will the price of Bitcoin..	YES	39.0%	28.0%	11.0%	7.6%	\$89	\$-89	\$1,077	L
3	Will the price of Etheru..	NO	47.9%	60.0%	12.1%	10.1%	\$109	\$+163	\$1,240	W
4	Will the price of Bitcoin..	YES	31.8%	0.2%	31.6%	15.8%	\$196	\$-196	\$1,044	L
5	Will the price of Bitcoin..	NO	50.0%	90.0%	40.0%	22.2%	\$232	\$-232	\$812	L
6	Will the price of Bitcoin..	NO	44.6%	56.5%	11.9%	10.5%	\$85	\$-85	\$727	L
7	Will the price of Etheru..	NO	51.3%	81.5%	30.2%	18.5%	\$135	\$-135	\$592	L
8	Will the price of Etheru..	YES	52.8%	24.0%	28.8%	18.9%	\$112	\$+355	\$947	W
9	Will the price of Bitcoin..	NO	51.3%	81.2%	29.9%	18.4%	\$174	\$-174	\$773	L
10	Will the price of Bitcoin..	NO	68.8%	94.0%	25.2%	13.4%	\$103	\$-103	\$669	L

Conclusions & Recommendations

Key Findings

1. Non-Crypto markets show strong, consistent alpha: +245% ROI, 61% win rate, Sharpe 0.27 over 38 trades.
2. Crypto predictions consistently underperform the market – the model's probability estimates for BTC/ETH price questions are systematically worse than market-implied probabilities. Win rate: ~30%.
3. Half-Kelly (0.5x) is near-optimal. Full Kelly over-bets and erases profits due to estimation noise. Quarter-Kelly (0.25x) is safer but leaves significant returns on the table.
4. All 9 aggregation mechanisms are profitable on non-crypto markets. Extremized performs best by ROI but with higher variance. Hybrid (M4) provides the best risk-adjusted performance.
5. Sports markets (NBA, NHL, CBB, Soccer) are the primary alpha source. The multi-expert debate structure surfaces analytical edges that single-model approaches and casual bettors miss.

Recommendations

1. DEPLOY on non-crypto markets only. Exclude all crypto price predictions.
2. USE half-Kelly (0.5x) position sizing with the Hybrid (M4) aggregator.
3. NO minimum edge filter needed – the Kelly criterion naturally sizes low-conviction bets smaller. Trading all signals maximizes total P&L.
4. SCALE UP data collection: 48 predictions over 7 days is promising but statistically limited. Target 200+ resolved predictions for significance.
5. INVESTIGATE crypto failure mode: the agents may lack real-time on-chain data and order flow information that crypto markets efficiently price in. Consider specialized crypto-only agents or excluding crypto entirely.

Caveats

- Small sample (48 predictions, 7-day window) – results may not generalize
- No slippage / liquidity constraints modeled
- Markets were pre-filtered (excluded extremes, geopolitical topics)